

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

SkyRide Operations Management System

Document Information

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Executive Summary

The SkyRide Operations Management System is a mobile application developed to support transportation personnel responsible for passenger pickup, onboarding, transportation, communication, payment collection, trip monitoring, and disruption management for tourism operations.

This document defines the detailed functional requirements required to deliver the business requirements outlined in the BRD.

Functional Scope

The application shall support:

- Authentication
- Dashboard Management
- Manifest Management
- Booking Management
- Passenger Management
- Communication Management
- Payment Management
- Route Management
- Trip Management
- Weather Cancellation Management
- Notifications
- Audit Logging

Module 1: Authentication

Description

Allows drivers to securely access the application.

Functional Requirements

FR ID	Requirement
FR-001	System shall allow login using registered username.
FR-002	System shall allow login using password.
FR-003	Username field shall be mandatory.
FR-004	Password field shall be mandatory.
FR-005	System shall validate credentials through authentication API.
FR-006	System shall display an error message for invalid credentials.
FR-007	System shall support Show Password functionality.
FR-008	System shall support Hide Password functionality.
FR-009	System shall redirect users to Dashboard after successful login.
FR-010	System shall automatically logout users after configured inactivity period.

Module 2: Forgot Password

Description

Allows drivers to recover access to their account.

Functional Requirements

FR ID	Requirement
FR-011	System shall provide Forgot Password option.
FR-012	System shall validate registered mobile number or email address.
FR-013	System shall generate password reset request.
FR-014	System shall send reset instructions to registered contact details.
FR-015	System shall display confirmation message after successful request.

Module 3: Dashboard

Description

Provides operational overview for drivers.

Functional Requirements

FR ID	Requirement
FR-016	System shall display assigned operations for the day.
FR-017	System shall display ride status.
FR-018	System shall display weather information.
FR-019	System shall display temperature information.
FR-020	System shall display active pickup manifests.
FR-021	System shall display trip progress.
FR-022	System shall enable dashboard cards based on ride status.

FR-023	System shall disable inaccessible actions.
FR-024	System shall navigate users based on operational status.

Module 4: Manifest Management

Description

Allows drivers to manage assigned passenger manifests.

Functional Requirements

FR ID	Requirement
FR-025	System shall display assigned pickup manifests.
FR-026	System shall display passenger counts.
FR-027	System shall display zones within manifest.
FR-028	System shall support passenger shuffling within zones.
FR-029	System shall support passenger shuffling between zones.
FR-030	System shall update manifest after successful shuffle.
FR-031	System shall support scrolling for large passenger lists.
FR-032	System shall display latest manifest information after refresh.

Module 5: Booking Management

Description

Allows drivers to view and manage booking information.

Functional Requirements

FR ID	Requirement
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FR-033	System shall display booking information.
FR-034	System shall identify incomplete passenger profiles.
FR-035	System shall identify incomplete payment records.
FR-036	System shall visually highlight incomplete bookings.
FR-037	System shall display booking status.
FR-038	System shall allow navigation to passenger details.

Module 6: Passenger Management

Description

Allows drivers to manage passenger onboarding and information.

Functional Requirements

FR ID	Requirement
FR-039	System shall display passenger details.
FR-040	System shall allow adding passengers.
FR-041	System shall allow adding child passengers.
FR-042	System shall allow updating child passengers.
FR-043	Child age shall not exceed 15 years.
FR-044	System shall validate mandatory passenger information.
FR-045	System shall display passenger onboarding status.
FR-046	System shall allow marking passenger as Boarded.
FR-047	System shall allow marking passenger as FTL.
FR-048	System shall update passenger status immediately.
FR-049	System shall prevent boarding if mandatory details are incomplete.
FR-050	System shall display warning popup before boarding incomplete passengers.

FR-051	System shall maintain passenger activity history.
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Module 7: Communication Management

Description

Allows communication with passengers.

Functional Requirements

FR ID	Requirement
FR-052	System shall provide Send Passenger Details Link option.
FR-053	System shall support Email communication.
FR-054	System shall support SMS communication.
FR-055	System shall display communication success confirmation.
FR-056	System shall display communication failure message.
FR-057	System shall maintain communication history.

Module 8: Payment Management

Description

Allows visibility and collection of passenger payments.

Functional Requirements

FR ID	Requirement
FR-058	System shall display payment status.
FR-059	System shall display outstanding balance.
FR-060	System shall generate payment link.
FR-061	System shall send payment link through SMS.

FR-062	System shall send payment link through Email.
FR-063	System shall update payment status after successful payment.
FR-064	System shall display payment confirmation.

Module 9: Route Management

Description

Allows drivers to execute assigned routes.

Functional Requirements

FR ID	Requirement
FR-065	System shall display assigned route.
FR-066	System shall display pickup stops.
FR-067	System shall display passenger pickup sequence.
FR-068	System shall allow trip initiation.
FR-069	System shall allow trip completion.
FR-070	System shall update route status in real time.

Module 10: Trip Status

Description

Provides visibility into active trip progress.

Functional Requirements

FR ID	Requirement
FR-071	System shall display current trip status.

FR-072	System shall display boarded passenger count.
FR-073	System shall display FTL passenger count.
FR-074	System shall display trip progress indicators.
FR-075	System shall update trip status automatically.

Module 11: Trip Summary

Description

Displays ride completion information.

Functional Requirements

FR ID	Requirement
FR-076	System shall generate trip summary.
FR-077	System shall display total passengers.
FR-078	System shall display boarded passengers.
FR-079	System shall display FTL passengers.
FR-080	System shall display completion timestamp.
FR-081	System shall allow users to review completed trips.

Module 12: Weather Cancellation

Description

Allows drivers to manage weather-related disruptions.

Functional Requirements

FR ID	Requirement
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FR-082	System shall notify users of weather disruptions.
FR-083	System shall display weather cancellation popup.
FR-084	System shall provide Cancel Operation option.
FR-085	System shall provide Rebook Operation option.
FR-086	System shall process cancellation requests.
FR-087	System shall process rebooking requests.
FR-088	System shall display confirmation messages.

Module 13: Notifications

Description

Handles system-generated notifications.

Functional Requirements

FR ID	Requirement
FR-089	System shall send operational notifications.
FR-090	System shall send payment notifications.
FR-091	System shall send passenger communication notifications.
FR-092	System shall log notification delivery status.

Module 14: Audit & Logging

Description

Tracks important system activities.

Functional Requirements

FR ID	Requirement
FR-093	System shall maintain login audit logs.
FR-094	System shall maintain passenger status change logs.
FR-095	System shall maintain payment activity logs.
FR-096	System shall maintain trip activity logs.
FR-097	System shall maintain weather cancellation logs.

Non-Functional Requirements

NFR ID	Requirement
NFR-001	Application response time shall be less than 3 seconds.
NFR-002	Application shall support Android devices.
NFR-003	Application shall use HTTPS for all communications.
NFR-004	Application shall maintain 99% uptime during operational hours.
NFR-005	Application shall support role-based access control.
NFR-006	Application shall securely store user credentials.
NFR-007	Application shall maintain audit logs for critical actions.
NFR-008	Application shall support concurrent users without degradation.
NFR-009	Application shall comply with data privacy standards.
NFR-010	Application shall provide disaster recovery capability.